

Blockchain-Assisted Privacy-Preserving Traffic Route Management Scheme for Fog-Based Vehicular Ad-Hoc Networks

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Abstract—Traffic route management is essential for reducing traffic jams and enhancing driving safety because of the growing number of vehicles and frequent occurrence of traffic accidents. However, in vehicular ad-hoc networks (VANETs), real-time messages are transmitted via wireless channels, which can result in security and privacy concerns. Existing proposals for traffic route management exist security vulnerabilities, as well as high calculation and communication costs. Encouraged by this fact, we design a lightweight traffic route management scheme for fog-based VANETs. In this scheme, vehicles utilize homomorphic encryption to encrypt their driving routes and then send the encrypted information to a fog node. The traffic management center (TMC) decrypts the received ciphertexts that are aggregated by the fog node and performs traffic management according to the decrypted data, without knowing individual route of each vehicle. Furthermore, blockchain is used in the scheme to conduct public keys management of vehicles. Our detailed security proof and analysis indicate that our proposal can meet the security objectives of VANETs. Further, to demonstrate the feasibility of the scheme, we also implement it in the Ethereum test network (i.e., *Rinkeby*). Significantly, the performance analysis demonstrates that our proposal achieves a better performance than other relevant representative schemes.

Index Terms—VANETs, privacy-preserving, traffic route management, homomorphic encryption, blockchain.

I. INTRODUCTION

OWING to the rapid growth in the number of vehicles and the continuous progress of communication technology,

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vehicular ad-hoc networks (VANETs) have become prominent research topics in both academia and industry recently. VANETs consist of three fundamental components for building an intelligent transportation system (ITS), such as vehicles, roadside units (RSUs), and trusted authorities (TAs). In VANETs, there are two main types of communication including vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) [1], [2]. In V2V communication, the vehicle periodically broadcasts traffic-related messages such as its speed, location, driving route, and so on, which can be used to enhance driving safety and reduce traffic accidents. In V2I communication, infrastructure such as an RSU is required to provide some traffic-related services to nearby vehicles. The leading communication technologies for realizing V2V and V2I communications are IEEE 802.11p and cellular networks [3].

Although VANETs have extensive applications and provide significant benefits, the prevailing security and privacy concerns cannot be ignored. VANETs are susceptible to multiple attacks owing to the public wireless networks used in V2V and V2I communication [4]. Without a secure protection mechanism for communication, messages transmitted in VANETs can be easily modified by the attackers [5]. For instance, driving route information can be altered maliciously to result in a traffic congestions, and speed information can be modified to result in a traffic accident. Therefore, traffic-related communications can affect personal life and property safety, it is extremely indispensable to design a message authentication scheme that is suitable for both V2V and V2I communications.

Because the number of vehicles is growing rapidly, the need for processing the associated data similarly expands. For example, traffic route aggregation and traffic sensing both require real-time traffic data produced by vehicles. Thus, to collect as well as process the data, cloud computing technology is introduced into the traditional VANETs architecture. However, cloud computing servers are far away from vehicles, which leads to great energy consumption and high latency [6], [7]. For overcoming these drawbacks, fog computing technology has been introduced to build the vehicular network models. In fog-based vehicular networks, fog nodes with certain computing and storage capabilities are distributed at the edge of the network, it can process the data that generated by vehicles more timely [8], [9]. A fog node can be any device, such as an RSU or a powerful server. The fog-based VANETs environments have several distinctive

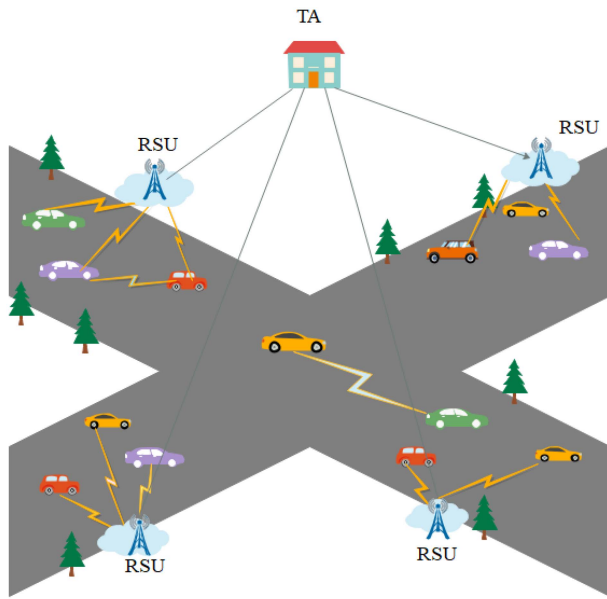


Fig. 1. Typical VANETs structure.

features, including low latency, location awareness, support for more edge nodes and wide geographical distribution [10].

In order to simultaneously achieve privacy and security, designing a privacy-preserving authentication mechanism is essential. During recent years, several various privacy-preserving authentication schemes have been proposed. These schemes can be divided into four types: public key infrastructure (PKI), identity-based, certificateless, and blockchain-based. We find that the PKI-based schemes are usually efficient and easy to implement, but the majority of them need great storage costs. Existing ID-based schemes have problems with key escrow and key revocation. In key escrow, the secret key of the vehicle is generated by the key generation center (KGC). However, if the KGC is compromised, the entire system is vulnerable to attack. Afterward, the key escrow problem has been solved using certificateless schemes, where the secret key of a vehicle is produced by both the KGC and itself. However, the key revocation problem remains unsolved in ID-based and certificateless schemes.

Blockchain technology has recently been introduced to the VANETs environment, and has drawn the attention of researchers in the realm of VANETs, because in the context of solving the above-mentioned security and performance problems, it affords the features of anonymity, decentralization, and immutability [11], [12]. Specifically, public information such as public keys, pseudonyms, and certificates can be managed by a smart contract in the blockchain, so that the authentication and revocation can be realized effectively. In addition, during authentication, it only needs to retrieve public information from the blockchain and not involve store new data into the blockchain.

To address the above-mentioned drawbacks, we design a blockchain-assisted privacy-preserving vehicle route management scheme for traffic management with a fog computing model, which is built by applying lightweight cryptographic technologies. In this scheme, each vehicle uses homomorphic

encryption to encrypt its driving routes. Next, it transmits the ciphertext to the fog node, which aggregates all vehicle routes. After aggregation, the fog node transmits the aggregated routes (not the route for each individual vehicle) to the TMC. Subsequently, the TMC decrypts the ciphertext, thus receiving the total number of vehicles in each segment without knowing their individual routes. The TMC can then conduct traffic management accordingly.

A. Our Contributions

The following are the main contributions of this paper that summarized by us.

- We propose a novel privacy-preserving vehicle route management scheme with a fog computing model based on elliptic-curve cryptography and homomorphic encryption algorithm. To guarantee the privacy of vehicle users, the TMC only knows the total number of vehicles in each segment without knowing the individual route of each vehicle. In addition, the scheme can support batch verification and is more appropriate for securing delay-sensitive applications in VANETs using a fog computing model.
- Blockchain is adopted to record the anonymous public keys of vehicles. Based on smart contract technology, it can perform update, query, and delete the public keys of vehicles that are recorded on the blockchain. In addition, during the authentication process, it does not require the trusted authority (TA) to be online, which can reduce the TA's workload.
- We provide a formal proof to indicate that the proposal is extremely secure and efficient. Meanwhile, the security analysis demonstrates that it can achieve several essential security objectives in VANETs. Additionally, the performance analysis reveals that the proposal can exhibit better performance than other related representative schemes.

B. Organization

Section II shows some related works. The preliminaries and background are demonstrated in Section III, including the system model, security objectives, and related preliminaries. We show the detailed process of our scheme in Section IV. The security proof and analysis are provided in Section V. Next, the performance analysis is introduced in Section VI. Finally, we draw the conclusion in Section VII.

II. RELATED WORK

A. Privacy and Security Research

To realize effective message authentication and ensure secure privacy-preserving mechanism in vehicular communication, lots of beneficial schemes have been proposed. In 2007, Raya and Hubaux [13] first designed a PKI-based scheme to achieve CPPA, in which a lot of anonymous public/private key pairs and the corresponding certificates are required to preload in vehicle's TPD, which brings about significant communication and storage overhead. Additionally, complex operations

are required to revoke the anonymous certificate. For solving the storage problems, Lu et al. [14] introduced an effective CPPA scheme where vehicles could ask for anonymous keys from their nearby RSUs. Even though it improves the storage efficiency of OBU, frequent interactions could result in a high authentication delay. Khodaei and Papadimitratos [15] proposed a practical scheme to distribute certificate revocation list in vehicle-centric system, which demonstrated that the scheme is highly efficient, scalable, and can withstand DoS attacks.

Since PKI-based schemes need to transmit digital certificates, the transmission delay and storage overhead may bring about a great effect on performance, especially when the number of vehicles is highly large. Thus, a lot of identity-based authentication schemes are designed over recent years. Zhang et al. [16] introduced an efficient identity-based batch signature verification scheme for VANETs. In their designed scheme, it can support batch verification and an RSU can verify multiple received signatures simultaneously, which can reduce the total verification time. Chim et al. indicated that many traditional schemes highly depended on a unhackable hardware device, which is unrealistic. Thus, they designed a software-based solution [17] which established two shared secrets with RSU and TA, and used the techniques of binary search and bloom filter to enhance the effectiveness of the verification phase. Liu et al. [18] introduced the concept of proxy vehicles to the VANETs. In their scheme, the proxy vehicles can authenticate multiple messages simultaneously, which can enhance authentication efficiency. Bagga et al. [2] proposed a bilinear pairing-based access control and key agreement scheme for smart transportation, it can ensure the secure communication among vehicles.

However, these related schemes are mostly pairing-based, which leads to high computational overhead. Therefore, some schemes are proposed without using bilinear pairing. In 2015, He et al. [19] firstly introduced an effective identity-based CPPA scheme, which is constructed on ECC rather than bilinear pairing. It has a better advantage in both communication cost and computation cost over other related schemes. Cui et al. [20] introduced a protocol on the basis of edge computing that used ECC cryptography, it supported batch verification with low redundancy authentication rate. Zhang et al. [21] proposed a CPPA scheme that is on the basis of ECC and Chinese remainder theorem in VANETs, their scheme settles the key leakage issue and the TPDs do not need to store system secret, which ensures the security of the whole system. Han et al. [22] designed an effective pairing-free aggregate signature scheme that is based on certificateless in VANETs, which supports batch verification and aggregate signature technique so that it can reduce the communication overhead.

B. Blockchain-Based Authentication Schemes

During recent years, lots of authentication schemes that are based on blockchain in VANETs have been proposed. To protect the privacy of vehicles, Lu et al. [23] designed an efficient blockchain-based scheme to achieve authentication for

vehicular communication. They utilize blockchain to record all certificates and transactions, and construct a new data structure named Merkle Patricia Tree to expand the traditional blockchain structure. However, the generation of anonymous certificates requires frequent interactions between vehicles and certification authorities. Lin et al. [24] designed a CPPA scheme that integrated blockchain technology and key derivation algorithm to manage certificates. But it required CA to update the public key of vehicles frequently, which lead to significant communication overhead and delay. Yao et al. [25] proposed a blockchain-assisted authentication scheme for vehicular fog environment. In their scheme, a vehicle can determine whether to be authenticated again or not when it enters a new space. Meanwhile, the scheme can eliminate the communication between service managers by adopting blockchain.

In 2020, Ma et al. [26] introduced an effective mechanism with blockchain technology to address key management issues for vehicular communications, which can achieve the vehicle public key's update, registration, and revocation automatically through smart contract. The cost of public key management of this scheme is much better than the traditional PKI-based management. Gabay et al. [27] proposed a authentication scheme to preserve the privacy of connected electric vehicles, which combined blockchain technology and zero-knowledge proofs. It eliminated the need for central authority and realized anonymous authentication, but they do not consider the traceability of malicious vehicles. In 2021, Yang et al. [28] designed a multi-domain CPPA authentication protocol that is based on blockchain to build distributed trust in VANETs. In this scheme, RSUs act as a proxy to assist in distributing pseudonyms, blockchain is regarded as a trust bridge to store cross-domain information, which can enhance the authentication efficiency. Later, Son et al. [29] designed a V2I handover authentication scheme that is based on blockchain for vehicular communication, which could utilize blockchain to reduce communication overhead during the second authentication and it supports revoking vehicles without the help of TA.

C. Fog-Based Authentication Schemes

In 2019, Cui et al. [30] proposed an efficient and safe road condition monitoring authentication scheme based on fog computing, the scheme is proposed by using a fog computing framework that supports mobility, low latency, and location awareness. Liu et al. [31] proposed a secure and efficient outsourcing computing scheme in vehicular fog computing, which performs outsourcing computing through fog vehicles with computing resources, and combines lightweight Boneh-Lynn-Shacham (BLS) signature and group signature to achieve batch anonymous authentication of fog vehicles while protecting their privacy in multiple outsourcing tasks. Cui et al. [30] proposed a privacy-preserving aggregation-authentication scheme for safety warning system in fog-cloud based VANETs. The scheme is realized using a novel efficient anonymous certificateless aggregation signcryption scheme, and allows a fog node to aggregate signcrypted traffic-related messages from surrounding vehicles into an

aggregated ciphertext and unencrypt them in a batch. Zhong et al. [32] proposed a secure and lightweight conditional privacy-preserving authentication scheme for fog-based vehicular ad-hoc networks, the scheme is proposed using a fog computing model that supports mobility, low latency, and location awareness.

III. PRELIMINARIES AND BACKGROUND

A. Elliptic Curve Cryptography

Let F_p be a finite field, which is determined by a prime number p . Let a set of elliptic curve points E over be defined by the equation: $y^2 = x^3 + ax + b \pmod{p}$, where $a, b \in F_p$. Let the point at infinity be O , then O and other points on E compose an additive elliptic curve group G with the order q and other generator P [33]. The elliptic curve group G has some attributes as follows:

- **Additive:** Let P and Q be two points of group G . If P differs from Q , then we can get $R = P + Q$ where R is the intersection of E and the straight line connecting P and Q . If $P = Q$, then $R = P + Q$. If $P = -Q$, then $P + Q = O$.
- **Scalar point multiplication:** Let $P \in G$ and $m \in \mathbb{Z}_q^*$, the scalar multiplication of E is defined as $m \cdot P = P + P + \dots + P$.
- **Elliptic curve discrete logarithm problem (ECDLP):** Given two points P and Q , where $Q = xP \in G$, $x \in \mathbb{Z}_q^*$, it is computational difficult for a probabilistic polynomial-time (PPT) adversary to calculate x .
- **Elliptic Curve Computational Diffie Hellman Problem (ECCDHP):** Given one tuple $(P, xP \in G, yP \in G)$ where $x, y \in \mathbb{Z}_q^*$, it is highly hard for any PPT adversary to compute xyP .

B. Homomorphic Encryption

Homomorphic encryption is a cryptography technology based on the computational complexity theory of mathematical problems. An output is obtained by processing the homomorphic encrypted data, then decrypting the output, the result is identical as that obtained by processing the unencrypted original data in the same way. In fact, homomorphic encryption is such an encryption function, which encrypts the plaintext by adding and multiplying on the ring, and the result is equivalent to the corresponding operation on the ciphertext after encryption. Because of this good feature, people can entrust a third party to process the data without disclosing the information. The main processes of homomorphic encryption include keys generation, encryption, and decryption [34]. There are some brief descriptions about it as follows:

- **Public/private keys generation:** Select two large prime numbers p and q randomly and gain their product $n = p \cdot q$. The least common multiple of $p - 1$ and $q - 1$ is denoted by λ . In addition, the function $L(x)$ is defined as $(x - 1)/n$. Then, $L(g^\lambda \pmod{n^2})$ is equal to $g^{\lambda(n)} \equiv 1 \pmod{n}$, $g \in \mathbb{Z}_{n^2}^*$ and it is reversible modulo n . Then, the public key for homomorphic encryption is (n, g) and the corresponding private key is (λ, μ) .

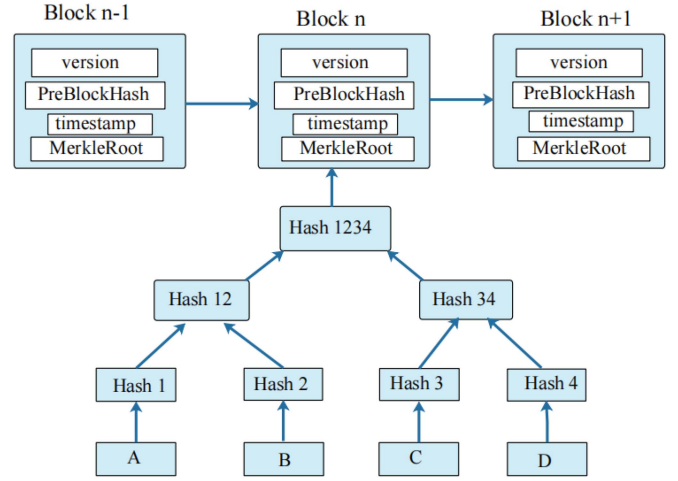


Fig. 2. Block structure of blockchain.

- **Encryption:** Given a message M , the ciphertext $C = E(M) = g^M \cdot r^n$ where $r \in \mathbb{Z}_{n^2}^*$.
- **Decryption:** Given the ciphertext C , the plaintext M is obtained by computing $M = D(C) = L(C^\lambda \pmod{n^2}) \cdot \mu \pmod{n}$.

C. Blockchain

As the basic technology of most existing cryptocurrencies, blockchain has attracted worldwide interest. Blockchain is actually equivalent to a disintermediated database, which is comprised of lots of data blocks [35], [36]. Each data block contains the information of a bitcoin network transaction, which is used to check the effectiveness of its information and generate the next block. Compared with the traditional network, blockchain has two core characteristics: one is difficult to tamper with the data, and the other is decentralized. Based on these two characteristics, the information recorded by blockchain is more reliable and can help solve the problem of people's distrust. Due to its characteristics, blockchain is becoming increasingly extensive, not just in the economic field [37]. The following is the block structure recorded on the blockchain.

A block in blockchain consists of a block header and a block body, where in the block header includes a hash of the previous block and a root of a Merkle tree. A large amount of complex data is recorded in the Merkle tree [38].

D. Smart Contracts and Ethereum platform

In the 1990s, Nick first proposed the concept of smart contract. It is a computer protocol with the characteristics of self-verifying, self-executing, and tamper-resistant. Each smart contract can be considered as a database with a unique address. Once the smart contract is affirmed by the consensus agreement and submitted to the blockchain, it will operate in a predetermined manner without interference from any third party. Complex interactions between different entities can be programmed into smart contracts, it can execute automatically immediately as soon as the conditions trigger the smart contract. It does not require human intervention and has the

Algorithm 1 Smart Contract: Update Function

Input: address user, uint256 txid
Output: bool result

```

1: require (SC.status is True);
3: if msg.sender is not TA then
4:   throw;
5: end
6: if Exist(VPKT[i].VPK==VPK) then;
7:   VPKT[i].VPK=newVPK;
8:   return True;
9: else
10:  return false;
11: end

```

Algorithm 2 Smart Contract: Query Function

Input: address user
Output: bool result

```

1: require (SC.status is True);
2: if Exist(VPKT[i].VPK==VPK) then;
3:   return True;
4: else
5:   return false;
6: end

```

Algorithm 3 Smart Contract: VoteUser Function

Input: address user
Output: bool result

```

1: require (SC.status is True);
3: if msg.sender is not fog node then
4:   throw;
5: end
6: if user is null then;
7:   return false;
8: else
9:   VoteCount+ = 1;
10:  return True;
11: end

```

advantage of flexibility [39], [40], [41], [42]. Ethereum is a platform that provides various modules for users to build applications, as a blockchain network, Ethereum is a decentralized public ledger used to verify and record transactions [43]. When a smart contract goes into effect, it cannot be interrupted or modified by any entity.

In our proposed scheme, we mainly utilize smart contract which can provide application binary interfaces (ABIs) to manage the vehicle public key table (VPKT). Namely, vehicles' public key is mapped to the transaction identities employing smart contract. Algorithms 1–4 demonstrate the smart contracts that provide different functions including updatePK, queryPK, VoteUser, and deletePK to manage the public key of vehicle.

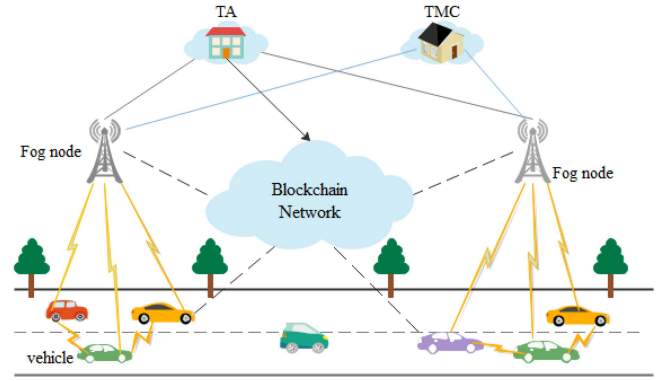


Fig. 3. System model.

Algorithm 4 Smart Contract: DeletePK Function

Input: address target
Output: bool result

```

1: require (SC.status is True);
3: if msg.sender is not TA then
4:   throw;
5: end
6: if target is null then;
7:   return false;
8: else if (VoteCount ≥ Thre) then
9:   delete VPKT[i].VPK;
10:  return True;
11: end

```

E. System Model

The involved communication entities in our scheme are illustrated in Figure 3, the corresponding responsibilities of each entity are defined as below.

- **TA:** The TA has a high ruling ability in the whole vehicular communication system. It possesses sufficient calculating, storage, and communication capability. Besides, it is in charge of the initialization of the whole system, as well as providing registration services for vehicles and fog nodes. Additionally, only the TA is capable of tracing the true identity of malicious vehicles. We suppose that the TA is a completely credible and uncompromisable entity.
- **Fog Node (FN):** Compared with the cloud, the fog node is distributed on the roadside with high calculating and storage power and is closer to the vehicle [44]. It is in charge of broadcasting route request messages, verifying vehicles' identities, and aggregating vehicles' driving routes. In addition, it can receive and analyze the messages and execute some decisions, such as real-time traffic warnings and traffic data aggregation. Moreover, we assume that FNs are honest and unwilling to leak any useful information in the scheme [14], [45].
- **Vehicle:** Vehicle acts as an extremely critical role in the intelligent transportation system. To improve traffic efficiency and driving safety, vehicles require to broadcast safety-related information including present velocity

and location to other vehicles or FNs periodically. Each vehicle is installed with an on-board unit (OBU) and a tamper-proof device (TPD). The OBU can make contact with other vehicles or infrastructures whereas the TPD can store security materials and perform the encryption and decryption operations [46]. Meanwhile, we assume that TPD is unhackable. Most of the vehicles are considered untrustworthy, so the identity of vehicles and information transmitted from vehicles require to be authenticated. Moreover, we assume that the vehicles have limited computing power.

- **TMC:** Each TMC is in charge of managing traffic conditions in its management area. In addition, it can guide the vehicles in time in case of traffic jams or traffic accidents. In our scheme, the TMC can decrypt the route aggregation content transmitted from the fog node and give timely route guidance to vehicles.

F. Security and Privacy Objectives

In VANETs, security and privacy concerns can not be ignored. The scheme should satisfy some security objectives as follows.

- 1) **Unlinkability:** The passive adversary does not have the ability to link the messages from the same vehicle successfully.
- 2) **Mutual authentication:** For ensuring the credibility of the communication process, the communication entities such as the vehicle, fog node, and TMC are able to authenticate each other for verifying the legitimacy of the identity and the integrity of the received message.
- 3) **Identity privacy-preserving:** To protect identity privacy, the vehicles use their pseudonym to communicate with other entities during the authentication process. Hence, the adversaries are unable to trace the true identity of vehicles through the received messages.
- 4) **Traceability:** When malicious vehicles send wrong traffic information to deceive others or deliberately transmit a false driving route so as to cause traffic jam, only the TA is capable of tracing the true identities of the malicious vehicles and performs some punishments.
- 5) **Resistance to replay attack:** It refers to that malicious adversaries send the same messages to the receiver repeatedly. Each message contains a timestamp, the receiver checks the validity of the received message. If the timestamp has terminated, the receiver will reject the message.
- 6) **Resistance to man-in-the-middle attack:** It means that the active attackers are unable to fabricate and modify the intercepted messages successfully.
- 7) **Resistance to common attacks:** The proposal is capable of withstanding some ordinary forms of attacks, e.g., offline password guessing attacks, modification attacks and impersonation attacks to guarantee the security of VANETs.

TABLE I
NOTATIONS

Notations	Definitions
TA	Trust authority
TMC	The traffic control center
V_i, F_j	i-th vehicle and j-th fog node, respectively
ID_i, ID_j	The real identity of vehicle and fog node, respectively
UID_i	The real identity of user
PW_i	The password of vehicle
s, P_{pub}	The private key and public key of the TA, respectively
sk_i, sk_j	The private key of vehicle and fog node, respectively
VPK, pk_j	The public key of vehicle and fog node, respectively
PID_i	The pseudonym of vehicle
T_{i1}, T_{j1}, T_{j2}	The latest timestamp
M_i	The message sent by vehicle and fog node
$\parallel$	Concatenation operation
$\oplus$	Exclusive-OR operation
E	An elliptic curve
G	The additive group with order q
P	The generator of the group G
h_i	secure one-way hash functions

IV. OUR PROPOSED SCHEME

In this part, we show a specific representation of the proposed scheme which comprises several steps. First of all, the TA initializes the entire system. Secondly, vehicles and fog nodes send their registration request to the TA, respectively. After completing this step, vehicle users need to pass the login phase successfully. Then, vehicles send their routes to the fog nodes, the fog nodes authenticate the legality of the vehicles, and then aggregate all routes received from vehicles. Next, the fog node transmits the aggregated ciphertext to the TMC, TMC decrypts it and conducts traffic management according to the decrypted ciphertext. At last, we provide a user-friendly password change step and revocation of malicious vehicles. The main notations and corresponding definitions are listed in Table I.

A. System Setup

At this stage, the TA prepares the essential system parameters, the specific descriptions are shown as follows.

- 1) The security parameter λ is selected as the input, the TA generates an elliptic curve $E : y^2 = x^3 + ax + b \pmod{p}$, where p is a random large prime number for the finite field F_p and $a, b \in F_p$. Then, the TA chooses a generator P to generate an additive elliptic curve group G with the order q .
- 2) The TA randomly chooses $s \in Z_q^*$ and calculates $P_{pub} = s \cdot P$, where s and P_{pub} are the system private key and system public key, respectively.
- 3) The TA chooses the following secure one-way hash functions: $h_1 : \{0, 1\}^* \times \{0, 1\}^* \rightarrow Z_q^*$, $h_2 : \{0, 1\}^* \times \{0, 1\}^* \times \{0, 1\}^* \rightarrow Z_q^*$, $h_3 : G \times \{0, 1\}^* \rightarrow Z_q^*$,

$h_4 : \{0,1\}^* \rightarrow Z_q^*$, $h_5 : \{0,1\}^* \times G \times \{0,1\}^* \times \{0,1\}^* \rightarrow Z_q^*$.

- 4) The TMC generates the public key (n, g) and private key (λ, μ) based on homomorphic encryption. It stores (λ, μ) secretly and transmits the public key (n, g) to the TA.
- 5) The TA broadcasts all public parameters and functions $\{p, q, a, b, P, G, P_{pub}, h_1, h_2, h_3, h_4, h_5\}$ via public channels.

B. Vehicle Registration

Before leaving the factory, each vehicle has to be registered with the TA through secure communication channels. After performing these steps, the vehicle obtains the private key and pseudo-identity from the TA. The detailed interaction processes are shown in the next description.

- 1) The vehicle user U_i chooses a password PW_i and his/her identity UID_i , then computes $UPW_i = h_1(UID_i \| PW_i)$. After that, the user sends registration request $\{ID_i, UPW_i\}$ to the TA via the reliable and private channel. To strengthen the safety of the scheme, the user encrypts the login password and its identity. Consequently, the attackers are unable to initiate internal attacks even though the TA is destroyed.
- 2) Once receiving the request message sent by the vehicle V_i , the TA first inspects whether V_i has already been registered or existed in the blacklist. If that is so, the TA refuses the request of V_i . Instead, the TA keeps executing the subsequent steps.
- 3) The TA randomly chooses $k_i \in Z_q^*$, and calculates $sk_i = h_2(ID_i \| s \| k_i)$ as V_i 's private key, the pseudonym of V_i is computed as $PID_i = ID_i \oplus h_3(sk_i \cdot P_{pub} \| t_i)$, where t_i is the latest timestamp. Then, the TA further computes $VPK = sk_i \cdot P$, $A_i = k_i P$, $B_i = A_i \oplus h_4(ID_i)$ and $C_i = h_3(A_i \| UPW_i)$.
- 4) The TA transmits $\{B_i, C_i, sk_i, PID_i, (n, g)\}$ to V_i and V_i stores them into its OBU secretly. In addition, the TA uploads $\{PID_i, VPK\}$ to the smart contract.

C. Fog Node Registration

This stage is executed in offline mode via reliable and private channels. After performing these steps, the fog node obtains a private key from the TA for the subsequent signature operation. The detailed interaction processes are shown in the next description.

- 1) The fog node F_j selects its identity ID_j and sends a registration request ID_j to the TA.
- 2) Once obtaining the registration request, the TA first checks whether the ID_j is available, if not, the registration request is refused by the TA. Instead, the TA randomly selects $sk_j \in Z_q^*$ and computes $PK_j = sk_j \cdot P$, where sk_j is the private key of F_j , PK_j is the public key of F_j .
- 3) Afterwards, the TA delivers $\{sk_j, pk_j\}$ to the fog node and publishes pk_j .
- 4) Lastly, the fog node F_j keeps sk_j secretly and finish the registration phase.

D. Login Phase

In this step, we aim to inspect the legality of the vehicle user.

- 1) The vehicle user U_i enters $\{ID_i, PW_i, UID_i\}$ into the OBU of vehicle V_i .
- 2) The V_i calculates $UPW_i = h_1(UID_i \| PW_i)$, $A_i = B_i \oplus h_4(ID_i)$, then checks if the equation $C_i = h_3(A_i \| UPW_i)$ holds or not. If it holds, this login request is permitted, then the vehicle V_i continues to the next phase. Otherwise, the login request is rejected by the V_i immediately.

E. Route Request From Fog Node

- 1) The fog node F_j chooses a number x_j randomly, and computes $X_j = x_j \cdot P$. Then, F_j generates the signature $\sigma_{F1} = sk_j + x_j \cdot h_2(ID_j \| RS \| T_{j1}) \mod q$, where T_{j1} is the latest timestamp, ID_j is the identity of F_j , RS is route request list that F_j want to query. After that, F_j broadcasts the message $M_1 = \{ID_j, X_j, \sigma_{F1}, RS, T_{j1}\}$ to all passing vehicles within its communication range.
- 2) Upon obtaining the message M_1 from F_j , V_i inspects the validity of timestamp T_{j1} firstly. If T_{j1} has terminated, V_i rejects the message M_1 . Instead, if T_{j1} is valid, V_i checks if the equation (1) holds.

$$\sigma_{F1} \cdot P = pk_j + X_j h_2(ID_j \| RS \| T_{j1}) \quad (1)$$

If it holds, V_i accepts the message and continues to the next step.

F. Route Reporting to Fog Node

After the vehicle V_i has determined the future driving routes, it requires to transmit the routes to the fog node. Owing to the personal driving routes are concerned with users' privacy, we need to encrypt their driving routes. Because the driving routes sent to the fog node are only used to get the aggregate package and the fog node cannot get the routes of each vehicle, homomorphic encryption is considered to be an appropriate method to encrypt the route that transmits to the fog node.

- 1) Firstly, V_i transforms the index of the route content that it intends to report into the binary format. The format of the driving route is denoted as the binary set $\{F_1(V), F_2(V), \dots, F_m(V)\}$, where the last bit of $F_i(V)$ indicates that V_i intends to drive on $NO.i$ route and the bit length of $F_i(V)$ equals to the base-2 logarithm of the max number of vehicles that one fog node can provide access services. For instance, $\{001, 000, 001, 000\}$ means that V_i will drive on $NO.1$ and $NO.3$ route.
- 2) Secondly, V_i randomly chooses $r_i \in Z_q^*$ and encrypts its route $\varphi_i = g^{F_0(i) \| F_1(i) \| \dots \| F_m(i)} \cdot r_i^n \mod n^2$. Then, V_i randomly selects $d_i \in Z_q^*$ and computes $D_i = d_i \cdot P$, $\beta_i = h_5(PID_i \| D_i \| \varphi_i \| T_{i1})$. Next, V_i generates a signature $\sigma_{Vi} = sk_i + d_i \cdot \beta_i \mod q$, where T_{i1} is the latest timestamp.
- 3) Last but not least, V_i transmits the message $M_{vi} = \{\sigma_{Vi}, \varphi_i, D_i, PID_i, T_{i1}\}$ to F_j .

G. Verification and Route Aggregation Phase

Once obtaining the message M_{vi} , F_j first inspects the validity of T_{i1} immediately. Only if T_{i1} is valid, a transaction is sent by the F_j to the blockchain for querying the public key. Upon querying the public key, the F_j computes $\beta_i = h_5(PID_i \| D_i \| \varphi_i \| T_{i1})$ for checking whether the equation (2) holds. Otherwise, the message is rejected by the F_j .

$$\sigma_{Vi} \cdot P = VPK + D_i \cdot h_5(PID_i \| D_i \| \varphi_i \| T_{i1}) \quad (2)$$

Due to $VPK = sk_i \cdot P$, $D_i = d_i \cdot P$, $\beta_i = h_5(PID_i \| D_i \| \varphi_i \| T_{i1})$, $\sigma_{Vi} = sk_i + d_i \cdot \beta_i \mod q$. Thus, we can gain that

$$\begin{aligned} \sigma_{Vi} \cdot P &= (sk_i + d_i \cdot \beta_i) \cdot P \\ &= VPK + d_i \cdot P \cdot \beta_i \\ &= VPK + D_i \cdot h_5(PID_i \| D_i \| \varphi_i \| T_{i1}) \end{aligned}$$

Batch Verification: Especially noteworthy is that our proposal can support batch verification. Namely, a fog node is able to verify multiple messages transmitted from others at the same time. To be specific, when it receives multiple messages $M_{vi} = \{\sigma_{Vi}, \varphi_i, D_i, PID_i, T_{i1}\}$ from other vehicles, where $i = 1, 2, \dots, n$, it executes the subsequent operations. Firstly, the fog node inspects the freshness of timestamp T_{i1} in each message, if the timestamp has terminated, then it discards the corresponding messages. To guarantee the integrity of signatures using batch verification, the small exponent test technology is introduced to the batch verification process [47]. Specifically, the fog node selects a vector $u = \{u_1, u_2, \dots, u_n\}$ randomly, where $u_i \in [1, 2^t]$ and t is a small integer. Then, the fog node queries the public key from the blockchain and computes $\beta_i = h_5(PID_i \| D_i \| \varphi_i \| T_{i1})$ respectively. At last, the fog node checks if equation (3) holds, if holds, it accepts the message tuples and continues to the next step. Otherwise, the message tuples are rejected by the fog node.

$$\begin{aligned} \left(\sum_{n=1}^n u_i \cdot \sigma_{Vi} \right) \cdot P &= \left(\sum_{n=1}^n u_i \cdot (sk_i + d_i \cdot \beta_i) \right) \\ &= \sum_{n=1}^n (u_i \cdot (sk_i \cdot P + \beta_i \cdot d_i \cdot P)) \\ &= \sum_{n=1}^n (u_i \cdot VPK + u_i \cdot \beta_i \cdot D_i) \\ &= \sum_{n=1}^n (u_i \cdot VPK) + \sum_{n=1}^n (u_i \cdot D_i \cdot \beta_i) \end{aligned} \quad (3)$$

After verifying the messages from vehicles, the fog node F_j aggregates the vehicles' routes and transmits the aggregated data to the TMC. In order to protect drivers' privacy, the TMC only knows the total number of vehicles to be driven in each section but cannot know the specific routes of each vehicle. The encrypted routes φ_{sum} are aggregated by the fog node as equation (4). Then, fog node F_j selects a number y_j randomly, computes $Y_j = y_j \cdot P$ and generates a signature

TABLE II
ROUTE AGGREGATION RESULT OF THE EXAMPLE

	$F_1(V)$	$F_2(V)$	$F_3(V)$	$F_4(V)$	$F_5(V)$
V0	000	001	001	000	001
V1	001	001	000	000	001
V2	000	000	001	001	001
V3	000	001	000	001	000
V4	001	001	001	000	001
V5	000	001	001	001	001
$\sum_{i=0}^5 V_i$	010	101	100	011	110
Number	2	5	4	3	6

$\sigma_{F2} = sk_j + y_j \cdot h_2(ID_j \| \varphi_{sum} \| T_{j2})$. After that F_j sends $M_3 = \{ID_j, \varphi_{sum}, \sigma_{F2}, Y_j, T_{j2}\}$ to the TMC.

$$\begin{aligned} \varphi_{sum} &= \prod_{i=0}^k \varphi_i \mod n^2 \\ &= \prod_{i=0}^k g^{F_0(i) \| F_1(i) \| \dots \| F_m(i)} \cdot r_i^n \mod n^2 \\ &= g^{\sum_{i=0}^k (F_0(i) \| F_1(i) \| \dots \| F_m(i))} \left(\prod_{i=0}^k r_i \right)^n \mod n^2 \\ &= g^{(\sum_{i=0}^k F_0(i) \| \sum_{i=0}^k F_1(i) \| \dots \| \sum_{i=0}^k F_m(i))} \left(\prod_{i=0}^k r_i \right)^n \mod n^2 \end{aligned} \quad (4)$$

H. Traffic Route Management Phase

Upon receiving $M_3 = \{ID_j, \varphi_{sum}, \sigma_{F2}, Y_j, T_{j2}\}$ from F_j , the TMC checks the timestamp T_{j1} firstly, and then verifies if the equation (5) holds. If not, TMC rejects the message. Instead, it performs the next steps.

$$\sigma_{F2} \cdot P = pk_j + Y_j \cdot h_2(ID_j \| \varphi_{sum} \| T_{j2}) \quad (5)$$

After that, the TMC utilizes the private (λ, μ) to decrypt the aggregated routes φ_{sum} by calculating $L(\varphi_{sum}^\lambda \mod n^2) \cdot \mu \mod n$, where $\mu = (L(g^\lambda \mod n^2))^{-1} \mod n$, $L(x) = (x - 1)/n$, and recovers the route information $RI = \sum_{i=0}^k F_0(i) \| \sum_{i=0}^k F_1(i) \| \dots \| \sum_{i=0}^k F_m(i)$.

To have a good understanding of our scheme, we take an instance to interpret the detailed process of the ultimate aggregated results and are ignorant of the detailed driving route of each vehicle. As shown in Table II assuming that six vehicles provide their future driving route, each route exists five sections and each section is indicated by three bits. The route provided by V_3 is 000 001 000 001 000, the TMC can get the total number $\sum_{i=0}^5 F_1(V) = 2, \dots, \sum_{i=0}^5 F_5(V) = 6$. Supposing that each section is assigned $nbits$ and the limitation of the total number of vehicles to provide route is no more than $2^n - 1$. After obtaining the route aggregation results, the TMC knows which segment is crowding and which segment has less vehicles, so as to has a correct and timely traffic guidance. To relieve traffic pressure, the TMC transmits traffic guidance informations to the fog node then broadcasts them to the vehicles. vehicles can decide whether to continue driving or change the another route through the traffic guidance message.

I. Vehicle Revoke

When the fog node discovers that vehicle V_i misbehaves, it encapsulates a voting transaction and transmits it to blockchain to trigger the smart contract. If vehicle V_i continues to carry out malicious behavior, the fog node adds a tag to *VoteCount*, once the value of *VoteCount* exceeds the set threshold *Thre*, i.e., $\text{VoteCount} \geq \text{Thre}$, a revocation transaction is sent by the TA to trigger smart contract *DeletePK* to remove the public key of the vehicle. Besides, TA checks the validity of the vehicle's public key periodically, if it has expired, TA also sends a transaction to trigger *DeletePK* to remove the vehicle's public key.

J. Password Change

A convenient password change procedure is provided by this scheme. The users are able to alter the login password at the vehicle side whenever they like. The specific descriptions are as follows.

- 1) The vehicle user U_i enters $\{UID_i, ID_i, PW_i\}$ and a new password PW_i^{new} into V_i .
- 2) V_i calculates $UPW_i = h_1(UID_i || PW_i)$, $A_i = B_i \oplus h_4(ID_i)$, and then checks if the equation $C_i = h_1(UID_i || PW_i)$ holds. If the password input by the user is incorrect, this step is ended. Otherwise, V_i computes $UPW_i^* = h_1(UID_i || PW_i^{new})$, $C_i^* = h_3(A_i || UPW_i^*)$ for changing PW_i into the new password PW_i^{new} .
- 3) Finally, V_i stores $\{B_i, C_i^*, UPW_i^*\}$ secretly.

V. SECURITY PROOF AND ANALYSIS

In this section, we firstly prove that the ECC-based signature in this scheme is existential unforgeability in the random oracle model. After that, a specific security analysis is conducted by us.

A. Formal Proof in the Random Oracle Model

Let $\mathcal{A}$ represent the adversary who is running against the scheme in polynomial time in the random oracle model. $\mathcal{A}$ is able to regulate the whole communication in vehicular networks. Specifically, $\mathcal{A}$ could modify, intercept and even forge informations. Let $\mathcal{C}$ represent the challenger who can settle the problems with a non-negligible probability by running $\mathcal{A}$ as a subroutine.

Theorem: If ECDLP problem is (τ', ϵ') -hard in G , then the signature scheme is (τ, ϵ) -existential unforgeable against adaptively choosen message attacks in the random oracle model [48], such that

$$\begin{aligned} \epsilon' &\geq \epsilon_1 \cdot \left(\frac{\epsilon_1}{q_{h2} + q_{h5}} + \frac{1}{q} \right) \\ \tau' &\leq 2\tau + (4t_s + 2t_a) \end{aligned} \quad (6)$$

where $\epsilon_1 = \epsilon - \frac{q_s(q_{h2} + q_{h5} + q_s)}{q}$, t_s and t_a indicate the time for scalar multiplication operation and point addition operation, respectively. In addition, q_{h2} , q_{h5} , q_s indicate the numbers of queries to h_2 oracle, h_5 oracle and sign oracle, respectively.

Proof: It is assumed that the success probability that adversary $\mathcal{A}$ against unforgeability of this scheme is denoted by ϵ .

$\mathcal{C}$ can solve the ECDLP with success probability ϵ' . Given a example $\{P, X = xP | P \in G, X \in G, x \in Z_q^*\}$, the target of $\mathcal{C}$ is output x . $\mathcal{C}$ maintains empty table L_{H2} and L_{H5} to simulate random oracles H_2 and H_5 . The following is $\mathcal{C}'$'s answers to $\mathcal{A}'$'s oracle queries.

Setup-query: The security parameter λ is selected as the input. Once receiving this query, $\mathcal{C}$ randomly chooses $s \in Z_q^*$ as its private key, then calculates the public key as $P_{pub} = s \cdot P$. After that, $\mathcal{C}$ transmits set $\{p, q, a, b, G, P_{pub}, P\}$ to $\mathcal{A}$.

h_2 -query: $\mathcal{C}$ generates a empty list L_{H2} . If $\mathcal{A}$ calls the query that uses set $\langle ID_i, s, k_i \rangle$, $\mathcal{C}$ inspects if the set has already existed in list L_{H2} . If exists, $\mathcal{C}$ will directly return the corresponding value to $\mathcal{A}$. If not, $\mathcal{C}$ randomly chooses $h_2 \in Z_q^*$, sets $h_2 = \langle ID_i, s, k_i \rangle$ and sends h_2 to $\mathcal{A}$.

h_5 -query: $\mathcal{C}$ produces a empty list L_{H5} . If $\mathcal{A}$ calls the query that uses the set $\langle PID_i, D_i, \varphi_i, T_{i1} \rangle$, $\mathcal{C}$ checks if the set has already existed in list L_{H5} . If exists, $\mathcal{C}$ will directly return the corresponding value to $\mathcal{A}$. If not, $\mathcal{C}$ randomly chooses $h_5 \in Z_q^*$, sets $h_5 = \langle PID_i, D_i, \varphi_i, T_{i1} \rangle$, and returns h_5 to $\mathcal{A}$.

Sign-query: The adversary $\mathcal{A}$ executes sign query in this step. After receiving the query with message $M_{vi} = \{PID_i, D_i, \varphi_i, T_{i1}\}$ from $\mathcal{A}$, $\mathcal{C}$ first selects number $s \in Z_q^*$, $d_i \in Z_q^*$, and $\sigma_{Vi} \in Z_q^*$ randomly, then computes $D_i = [\sigma_{Vi} \cdot P - h_2 \cdot P] \cdot \beta_i^{-1}$. If $h_2(ID_i, s, k_i)$ has already been defined, if so, $\mathcal{C}$ halts, returns $\perp$ and sets $bad \leftarrow true$. If $h_5(PID_i, D_i, \varphi_i, T_{i1})$ also has been defined, $\mathcal{C}$ halts, returns $\perp$ and sets $bad \leftarrow true$. Otherwise, $\mathcal{C}$ returns $\langle \sigma_{Vi}, D_i \rangle$ to $\mathcal{A}$. It is distinct that these simulations is indistinguish from the real scene.

After performing the above queries, we supposed that $\mathcal{A}$ outputs a forged signature $\langle \sigma_{Vi}, D_i \rangle$ on the message $\langle PID_i, D_i, \varphi_i, T_{i1} \rangle$. The forgery is no-trivial if $\mathcal{A}$ do not make a *Sign-query* on the $\langle PID_i, D_i, \varphi_i, T_{i1} \rangle$.

Let $E1$ represent the event that $\mathcal{C}$ does not abort due to the signature simulation, $E2$ represent the event that $\mathcal{A}$ returns a non-trivial forgery. The probability of $\mathcal{A}$ forging a signature $\langle \sigma_{Vi}, D_i \rangle$ on message $\langle PID_i, D_i, \varphi_i, T_{i1} \rangle$ is $\epsilon_1 = Pr[E1]Pr[E2|E1]$. The following are the calculation processes.

$$\text{Claim 1: } Pr[E1] = Pr[\neg bad] \geq 1 - \frac{q_s(q_{h2} + q_{h5} + q_s)}{q}$$

Proof: We have $bad \leftarrow true$ if the pair $\langle ID_i, s, k_i \rangle$ generated in a *Sign* simulation has been occurred by coincidence in a former query to the h_2 query, and if $\langle PID_i, D_i, \varphi_i, T_{i1} \rangle$ generated in a *Sign* simulation has been occurred in a former query to h_5 oracle. Because there are $q_{h2} + q_{h5} + q_s$ entries at most, the probability of this event is $\frac{q_s(q_{h2} + q_{h5} + q_s)}{q}$ at most for one *Sign-query*. Consequently, the probability of this event for q_s queries is $\frac{q_s(q_{h2} + q_{h5} + q_s)}{q}$ at most.

$$\text{Claim 2: } Pr[E2|E1] \geq \epsilon$$

Proof: The value of $Pr[E2|E1]$ is the probability that adversary $\mathcal{A}$ sends a valid forgery supposing that $\mathcal{C}$ does not abort due to $\mathcal{C}'$'s *Sign* queries. If $\mathcal{C}$ did not abort due to $\mathcal{A}'$'s queries, all its replies to these queries are valid. Hence, according to the hypothesis, $\mathcal{A}$ will generate a non-trivial forgery with probability ϵ at least.

Thus, the probability that $\mathcal{A}$ sends a valid forgery is $\epsilon - \frac{q_s(q_{h2} + q_{h5} + q_s)}{q}$ at least. Afterwards, $\mathcal{C}$ performs Forking

lemma algorithm of Bellare and Neven [49] to get two valid forgeries σ_{V_i} and σ'_{V_i} as shown in equation (7).

$$\begin{aligned}\sigma_{V_i} &= sk_i + d_i \cdot h_5 \bmod q \\ \sigma'_{V_i} &= sk_i + d_i \cdot h'_5 \bmod q\end{aligned}\quad (7)$$

where $\sigma_{V_i} \neq \sigma'_{V_i}$, $h_5 \neq h'_5$. Therefore, the solution to ECDLP, sk_i , is calculated as presented in equation (8).

$$sk_i = \frac{h'_5 \sigma_{V_i} - h_5 \sigma'_{V_i}}{h'_5 - h_5} \bmod q \quad (8)$$

On the basis of Forking lemma of Bellare and Neven [49] that $frk \geq acc \cdot (\frac{acc}{q} + \frac{1}{h})$, the C' 's probability of success is:

$$\epsilon' \geq \left(\epsilon - \frac{q_s(q_{h2} + q_{h5} + q_s)}{q} \right) \cdot \left(\frac{\epsilon - \frac{q_s(q_{h2} + q_{h5} + q_s)}{q}}{q_{h2} + q_{h5}} + \frac{1}{q} \right) \quad (9)$$

In order to compute the ECDLP issue, provided that a scalar multiplication operation in G takes time t_s , point addition operation in G takes time t_a , while the time of all other operations is not taken into account. The running time of C is twice that of A , plus the time which is demanded to respond to hash queries and sign queries. Thus, the running time of C is $\tau' \leq 2\tau + q_s(4t_s + 2t_a)$. Hence, we complete the security proof.

B. Informal Security Analysis

- 1) *Unlinkability*: Each pseudo-identity PID_i of vehicle V_i and messages transmitted over the vehicular network are encrypted by timestamp and random number, due to the random number and the timestamp are random, so the pseudonym will be updated dynamically and each message is different. Therefore, the adversary cannot link various messages to the identical vehicle successfully.
- 2) *Mutual authentication*: In this scheme, vehicle and the fog node can authenticate the received messages M_1 and M_{vi} by verifying whether their obtained messages are valid or not respectively. Besides, the TMC can authenticate the message M_3 transmitted from the fog node. Because hash function and the ECDLP are able to resist collision, any adversary cannot produce a valid message in polynomial time. Therefore, mutual authentication can be realized in this proposal.
- 3) *Identity privacy-preserving*: In this proposed scheme, the pseudonym is used by V_i to transmit messages instead of using the real identity, and the TA updates pseudonym PID_i of V_i dynamically for each communication, where $PID_i = ID_i \oplus h_3(sk_i \cdot P_{pub} \| t_i)$, $sk_i = h_2(ID_i \| s \| d_i)$. Consequently, the attackers are unable to disclose the true identity of V_i because of the ECDLP.
- 4) *Traceability*: When the vehicle misbehaves, the TA is capable of retrieving the real identity of the malicious vehicle by calculating $ID_i = PID_i \oplus h_3(sk_i \cdot s \cdot P \| t_i) = PID_i \oplus h_3(VPK_i \cdot s \| t_i)$, where s is the system private key. Therefore, this scheme could support traceability.
- 5) *Resistance to replay attack*: To withstand replay attacks, the freshness of the message is required to inspect by the

message receiver firstly. Each message sent from V_i and F_j contains a timestamp t_{ii} . For the sake of inspecting the timestamp of message M , a receiver computes and verifies whether the equation $|t_{ii} - t_{cur}| \leq \Delta t$ is true. If the timestamp has expired, the receiver will reject the message.

- 6) *Resistance to common attacks*: In addition to the above-mentioned attacks, the proposal is capable of withstanding the following attacks.

- *Offline password guessing attack*: During the registration, the password PW_i is encrypted, where $UPW_i = h_1(UID_i \| PW_i)$, $C_i = h_3(A_i \| UPW_i)$. In addition, the password PW_i can be altered by legal users frequently. Thus, attackers cannot infer the login password successfully.
- *Modification attack*: Resistance to modification attack means that the active adversary can not modify a message to cheat the message receiver. In our proposed scheme, the interacted messages mainly involve $M_1 = \{ID_j, X_j, \sigma_{F1}, RS, T_{j1}\}$, $M_{vi} = \{\sigma_{V_i}, \varphi_i, D_i, PID_i, T_{i1}\}$ to F_j and $M_3 = \{ID_j, \varphi_{sum}, \sigma_{F2}, Y_j, T_{j2}\}$. The active adversary can not modify these messages successfully because the modified messages cannot be successfully verified. Thus, the scheme can resist modification attacks.
- *Impersonation attack*: Our proposed scheme could withstand impersonation attacks as follows:
Case 1 (Vehicle Impersonation Attack): We assume that the legitimate message M_{vi} transmitted by the vehicle V_i is seized by the adversary A , the adversary A attempts to get beneficial information from the M_{vi} and to produce a valid message to cheat the authentication of F_j . However, the A does not know the sk_i , s , d_i and T_{i1} which exists in the M_1 . Thus, A does not have the ability to generate valid message M_{vi} in polynomial time.
Case 2 (Fog Node Impersonation Attack): We assume that the valid message M_3 sent by the fog node F_j is seized by the adversary A , the adversary A attempts to obtain beneficial information from the M_3 and to produce a valid message to cheat the authentication of TMC. However, the A does not know the ID_j , sk_j , y_j , T_{j2} contained in M_3 . Therefore, the adversary A does not have the ability to generate valid message M_3 in polynomial time.

VI. PERFORMANCE ANALYSIS

In this section, we analyze the performance of our proposal during the authentication process in terms of communication and calculation cost. Then we give comparisons against existing scheme [50], [51] and [52], where Shen et al.'s [50] and Rabieh et al.'s [51] are based on bilinear pairing operations, Cui et al.'s [52] and our scheme are based on ECC.

In order to evaluate computation and communication overhead as well as realize an 80-bit security level, we construct a bilinear pairing $e : \mathbb{G}_1 \times \mathbb{G}_1 \rightarrow \mathbb{G}_T$, where $\mathbb{G}_1$ with the

TABLE III
THE EXECUTION TIME OF BASIC CRYPTOGRAPHIC OPERATIONS (MS)

Operations	Execution time (ms)
T_{ecc-sm}	0.3218
T_{ecc-pa}	0.0024
T_h	0.001
T_{mtp}	0.0992
T_{bp}	5.086
T_{bp-sm}	0.694
T_{bp-pa}	0.0018
T_e	2.126
T_s	0.276

order $\bar{q}$ is a additive group generated by $\bar{P}$ over the super singular elliptic curve $\bar{E} : y^2 = x^3 + x \bmod \bar{p}$ with embedding degree 2. $\bar{p}$ and $\bar{q}$ are a 512-bit prime number and a 160-bit prime number, respectively. Simultaneously, we construct an additive group $\mathbb{G}$ with order q that generated by P over a non-singular elliptic curve $E : y^2 = x^3 + ax + b \bmod p$, where $a, b \in Z_q^*$, p, q are 160-bit prime numbers.

A. Computational Overhead Analysis

MIRACL library [53] is used to measure the execution times of involved cryptographic operations. The experiment is tested on a personal computer under Windows 10 environment, which is equipped with Ubuntu 16.04, i5-MX350 CPU, and 4G RAM. Table III reveals the execution time of related cryptography operations. Owing to the time of XOR operation can be ignored, we do not take this operation time into consideration. For convenience, some symbols about execution time are defined as shown below.

- T_{ecc-sm} : The time for executing a scale multiplication operation $x \cdot P$ that associated with the ECC, where $x \in Z_q^*$ and $P \in \mathbb{G}$.
- T_{ecc-pa} : The time for executing a point addition operation $P + Q$ that associated with the ECC, where $P, Q \in \mathbb{G}$.
- T_h : The time for executing a one-way hash function operation.
- T_{bp} : The time for performing a bilinear pairing operation $\bar{e}(P, Q)$, where $P, Q \in \mathbb{G}_1$.
- T_{mtp} : The time for executing a MapToPoint hash operation associated with the bilinear pairing.
- T_{bp-sm} : The time for executing a scale multiplication operation $\bar{x} \cdot \bar{P}$ that associated with the bilinear pairing, where $\bar{x} \in Z_q^*$, $P \in \mathbb{G}_1$.
- T_{bp-pa} : The time for executing a point addition operation $\bar{P} + \bar{Q}$ that associated with the bilinear pairing, where $\bar{P}, \bar{Q} \in \mathbb{G}_1$.
- T_e : The time for executing an exponentiation operation over $\mathbb{G}$.
- T_s : The time for executing an symmetric encryption/decryption operation.

Next, we analyze the computational overhead generated in related operations performed in the authentication processes of our scheme and schemes [50], [51], and [52]. In scheme [50], a vehicle requires to execute one modular exponentiation operation, one multiplication operation, one point addition

operation, one bilinear pairing operation, and three hash operations during the authentication process. Consequently, the computational overhead on the vehicle terminal is $T_e + T_{bp-sm} + T_{bp} + T_{bp-pa} + 3T_h \approx 7.9108ms$. During the authentication process, the RSU is required to perform two modular exponentiation operations, three multiplication operations, two bilinear pairing operations, two point addition operations, and three hash operations. Therefore, the overhead needed on the RSU is $2T_e + 3T_{bp-sm} + 2T_{bp} + 2T_{bp-pa} + 3T_h \approx 16.5126ms$. So the total time of [50] is $3T_e + 4T_{bp-sm} + 3T_{bp} + 3T_{bp-pa} + 6T_h \approx 24.42ms$.

In Rabieh et al.'s scheme [51], a vehicle requires to execute one exponentiation operation, one multiplication operation, two bilinear pairing operations, and two hash operations. Hence, the computational overhead on the vehicle terminal is $T_e + T_{bp-sm} + 2T_{bp} + 2T_{mtp} \approx 13.19ms$. The computational overhead needed on RSU side includes one exponentiation operation, two multiplication operations, two bilinear pairing operations, and three hash operations. Thus, the execution time is $T_e + T_{bp-sm} + 2T_{bp} + 3T_{mtp} \approx 13.29ms$. TMC needs to execute one exponentiation operation, two bilinear pairing operations, and one hash operation. Namely, the computational overhead is $T_e + 2T_{bp} + T_{mtp} \approx 12.40ms$. Thus, the total computational overhead of [51] is $3T_e + 2T_{bp-sm} + 5T_{bp} + 6T_{mtp} \approx 38.88ms$. In [52], the computational overhead on the vehicle side is $T_e + 5T_{ecc-sm} + 7T_h + T_s \approx 4.02ms$. On the RSU side, the computational overhead is $T_e + 3T_{ecc-sm} + 2T_h + T_{ecc-ap} \approx 3.096ms$. The computational overhead on the CSP is $6T_{ecc-sm} + 2T_{ecc-pa} + T_s + 6T_h \approx 2.218ms$. Hence, the total computational overhead is $2T_e + 14T_{ecc-sm} + 15T_h + 2T_s + 3T_{ecc-pa} \approx 9.334ms$.

In this proposal, a vehicle needs to execute one exponentiation operation, three multiplication operations, one point addition operation, two hash operations. Thus, the execution time on the vehicle side is $T_e + 3T_{ecc-sm} + T_{ecc-pa} + 2T_h \approx 3.096ms$. On the fog node side, it requires one exponentiation operation, five multiplication operations, one point addition operation, and three hash operations, so the time is $T_e + 4T_{ecc-sm} + T_{ecc-pa} + 3T_h \approx 3.42ms$. The TMC needs to execute one exponentiation operation, two multiplication operations, one point addition operation, and one hash operation. So the execution time on the TMC side is $T_e + 2T_{ecc-sm} + T_{ecc-pa} + T_h \approx 2.773ms$. Consequently, the total cost of this scheme is $3T_e + 9T_{ecc-sm} + 3T_{ecc-pa} + 6T_h \approx 9.29ms$.

B. Communication Overhead Analysis

Owing to p is 20 bytes and $\bar{p}$ is 64 bytes, the elements in $\mathbb{G}$ and $\mathbb{G}_1$ are $20 \times 2 = 40$ bytes and $64 \times 2 = 128$ bytes, respectively. Besides, let the timestamp and identity be 4 bytes as well as the output of general hash function be 20 bytes. We take the size of messages in the course of the authentication phase into account only.

In our scheme, there are three rounds during the authentication process, the transmitted messages including $M_1 = \{ID_j, X_j, \sigma_{F1}, RS, T_{j1}\}$, $M_{vi} = \{\sigma_{Vi}, \varphi_i, D_i, PID_i, T_{i1}\}$, and $M_3 = \{ID_j, \varphi_{sum}, \sigma_{F2}, Y_j, T_{j2}\}$. Due to X_i, D_i , and

TABLE IV
COMPUTATION OVERHEAD (MS)

	Vehicle	RSU/FN	TA/CSP/TMC	Total
Shen et al [50].	$T_e + T_{bp-sm} + T_{bp} +$ $T_{bp-pa} + 3T_h \approx 7.9108$	$2T_e + 3T_{bp-sm} + 2T_{bp} +$ $3T_h \approx 16.5126$	/	$3T_e + 4T_{bp-sm} + 3T_{bp-pa} +$ $6T_h \approx 24.42$
Khaled et al [51].	$T_e + T_{bp-sm} + 2T_{bp}$ $+2T_{mtp} \approx 13.19$	$T_e + T_{bp-sm} + 2T_{bp}$ $+3T_{mtp} \approx 13.29$	$T_e + 2T_{bp} + T_{mtp} \approx 12.40$	$3T_e + 2T_{bp-sm} + 5T_{bp}$ $+6T_{mtp} = 38.88$
Cui et al [52].	$T_e + 5T_{ecc-sm} + 7T_h +$ $T_s \approx 4.02$	$T_e + 3T_{ecc-sm} + 2T_h$ $+T_{ecc-pa} \approx 3.096$	$6T_{ecc-sm} + 2T_{ecc-pa}$ $+T_s + 6T_h \approx 2.218$	$2T_e + 14T_{ecc-sm} + 15T_h$ $+2T_s + 3T_{ecc-pa} \approx 9.334$
Our scheme	$T_e + 3T_{ecc-sm} + T_{ecc-pa}$ $+2T_h \approx 3.096$	$T_e + 5T_{ecc-sm} + T_{ecc-pa}$ $+3T_h \approx 3.42$	$T_e + 2T_{ecc-sm} + T_{ecc-pa}$ $+T_h \approx 2.773$	$3T_e + 9T_{ecc-sm} + 3T_{ecc-pa}$ $+6T_h \approx 9.29$

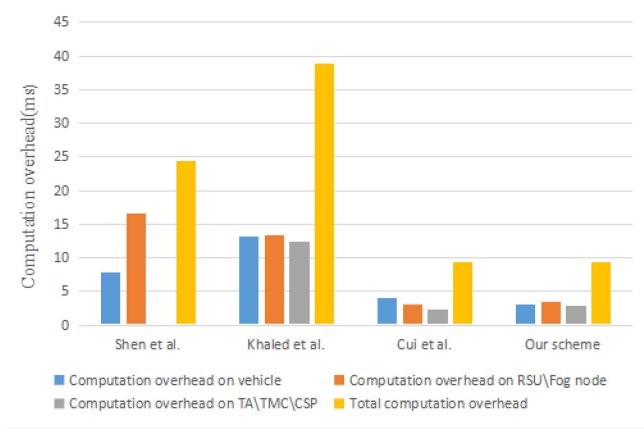


Fig. 4. Computation overhead comparison.

Y_j belong to $\mathbb{G}$, ID_j belongs to identity database, T_{j1} , T_{i1} , and T_{j2} are the timestamps, $PID_i, \sigma_{F1}, \sigma_{Vi}, \sigma_{F2}$ are the result of regular hash operation, φ_i and φ_{sum} are the result of modular exponential operation. Thus, the communication overhead of the scheme is $|M_1| = 4 + 40 + 20 + 4 + 4 = 72$ bytes, $|M_{Vi}| = 20 + 128 + 40 + 20 + 4 = 212$ bytes, $|M_3| = 4 + 128 + 20 + 40 + 4 = 196$ bytes, respectively. Consequently, the total communication overhead is $72 + 212 + 196 = 380$ bytes.

In Shen's scheme [50], a vehicle requires to send $M_1 = \{ID_i, T_i, \sigma_i = (e(g, g)^{r1}, r_2, \eta)\}$, because ID_i belongs to identity databases, T_i is timestamp, $\eta \in \mathbb{G}_1$, $e(g, g)^{r1} \in \mathbb{G}_T$, r_2 is the result of general hash operation. So the length of M_1 is $|M_1| = 4 + 4 + 20 + 128 + 256 = 412$ bytes. A RSU also needs to send M_1 to the TCS, so the total communication overhead is $412 + 412 = 824$ bytes. The detailed communication overhead analysis of Rabieh's scheme [51] and Cui's scheme [52] are similar to the above. The total communication overhead of [52] and [51] are 592 bytes and 340 bytes, respectively.

C. Blockchain Practical Feasibility Analysis

For discussing the practicability of the proposal, we implement it on *Rinkeby* (which is an Ethereum test network) to

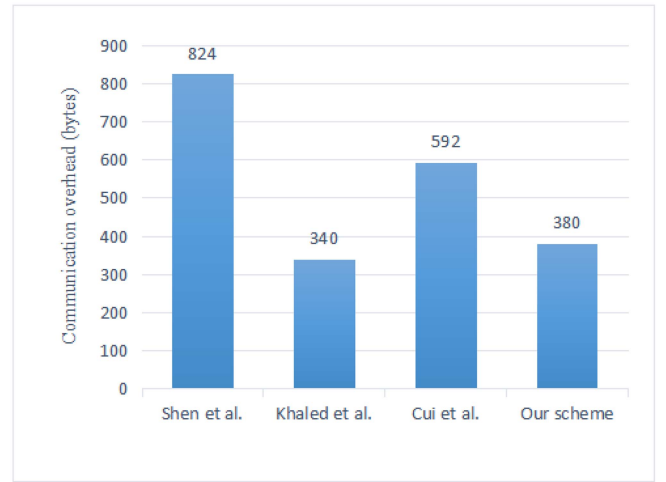


Fig. 5. Communication overhead comparison.

deploy and invoke the designed smart contract. When the written solidity smart contract code has been compiled, we deploy it on *Rinkeby* to test the gas cost of all operations. The cost of gas is one of the relevant factors measured in Ethereum, which is a reflection of the cost that uses blockchain [54]. The detailed descriptions of this implementation are shown as follows.

First of all, to test the cost, we employ MetaMask to produce two accounts (TA and fog node), which the addresses are `0xB151ff018fC32fAb28E85E521BC071fb1c896bA8` and `0x95A10EBC57D0A183bea52d7DBD692A3edF7E5303`, respectively. Then, we switch to the TA's account to request two Ethers from the *Rinkeby* test network, so that the TA can publish related transactions to update, revoke the public key of vehicle. When the transaction is recorded into the *Rinkeby*, others can retrieve it from the blockchain.

Next, we use Remix as fog node to deploy the smart contract into *Rinkeby* and get its address (i.e., `0x95A10EBC57D0A183bea52d7DBD692A3edF7E5303`). We also invoke the query algorithm to query the public key of vehicle from blockchain.

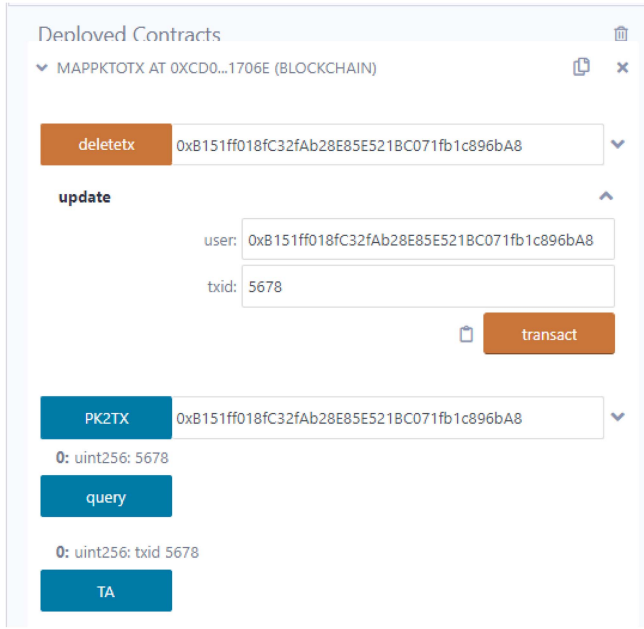


Fig. 6. Deploy smart contract.



Fig. 7. Execute the update function.



Fig. 8. Execute the delete function.

What's more, for testing the transaction cost, we evaluate the gas cost of these operations, and the results are presented in Table V. The price on June 17, 2022 is 1 ETH = 1075 USD, the GAS PRICE is set to 2 Gwei (1 Gwei = 10^{-9} ether). From TABLE V, we can see that deploying smart contract is the maximum cost, which is about USD 0.9284, it is just

TABLE V
GAS COST OF SMART CONTRACT (GAS PRICE=2 GWEI, 1
ETHER=1075 USD)

Operation	Gas Used	Actual Cost (ether)	USD
deploy	431800	0.000863600	0.9284
update	59375	0.000118750	0.1277
query	25742	0.000051484	0.0553
delete	37580	0.000075160	0.0808

implemented once. While all other related operations are called frequently, the cost of them is not more than USD 0.13 which is within acceptable range.

VII. CONCLUSION

In this study, to reduce traffic jams and ensure secure communication in VANETs, we designed a lightweight privacy-preserving traffic route management scheme that incorporates blockchain technology and a fog computing model. In the proposed scheme, the TMC only knows the total number of vehicles in each segment, not the individual routes of a vehicle. In order to realize efficient vehicular communication, fog computing was introduced to support mobility, low latency, and location awareness. Furthermore, our scheme can revoke illegal vehicles dynamically through smart contracts and it can assist the fog nodes to batch verify multiple messages from vehicles. Finally, the formal security proof and specific security analysis demonstrate that the proposal can satisfy fundamental security and privacy objectives of VANETs, and is able to resist common types of attacks. Meanwhile, our performance analysis indicates that the scheme has a certain advantage over other related representative schemes, and is suitable for fog-based VANETs, especially for applications that are sensitive to time delay.

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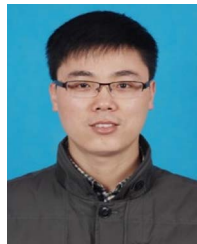


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